

- 6 (a)** hadron not a fundamental particle/lepton is fundamental particle
 or
 hadron made of quarks/lepton not made of quarks
 or
 strong force/interaction acts on hadrons/does not act on leptons B1 [1]
- (b) (i)** ${}^0_1\text{e}^{(+)}$ or ${}^0_1\beta^{(+)}$ B1
- ${}^0_0\nu_{(\text{e})}$ B1 [2]
- (ii)** weak (nuclear force / interaction) B1 [1]
- (iii)**
- mass-energy
 - momentum
 - proton number
 - nucleon number
 - charge
- Any three of the above quantities, 1 mark each* B3 [3]
- (c)** (quark structure of proton is) up, up, down or uud B1
- up/u (quark charge) is $(+) \frac{2}{3}(e)$, down/d (quark charge) is $- \frac{1}{3}(e)$ C1
- $\frac{2}{3}e + \frac{2}{3}e - \frac{1}{3}e = (+)e$ A1 [3]